Problem 3

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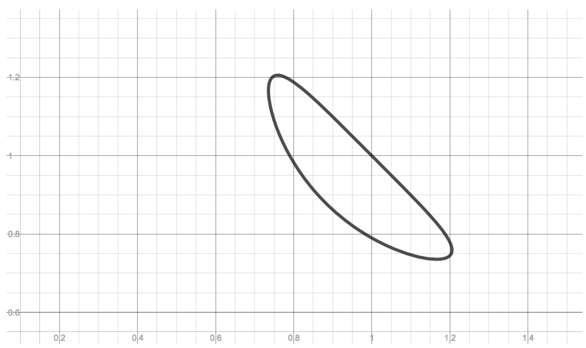
Problem 1 *A part of the curve with equation $\cos(\pi xy)+x+y = 1$ is sketched below.*



(a) *Use the implicit differentiation to find the derivative dy/dx.*

Explanation. Implicitly differentiate equation:

$$\frac{d}{dx}(\cos(\pi xy) + x + y) = \frac{d}{dx}(1) \implies -\sin(\pi xy) \cdot (\pi y + \pi x\frac{dy}{dx}) + 1 + \frac{dy}{dx} = 0$$

Solve for $\dfrac{dy}{dx}$:

$$\begin{aligned}
-\sin(\pi xy) \cdot (\pi y + \pi x\frac{dy}{dx}) + 1 + \frac{dy}{dx} = 0 &\implies -\pi y\sin(\pi xy) - \pi x\frac{dy}{dx}\sin(\pi xy) + 1 + \frac{dy}{dx} = 0 \\
&\implies \big(1 - \pi x\sin(\pi xy)\big)\frac{dy}{dx} = \pi y\sin(\pi xy) - 1 \\
&\implies \frac{dy}{dx} = \frac{\pi y\sin(\pi xy) - 1}{1 - \pi x\sin(\pi xy)} \qquad \text{if } 1 - \pi x\sin(\pi xy) \neq 0
\end{aligned}$$

(b) *Consider the point $(1,1)$. Show (algebraically) that this point lies on the curve.*

Explanation. $(1,1)$ lies on this curve:

$$\begin{aligned}
\cos(\pi \cdot 1 \cdot 1) + 1 + 1 &= \cos(\pi) + 2 \\
&= -1 + 2 = 1
\end{aligned}$$

(c) *Find the equation of the line tangent to the curve at* $(1, 1)$. *Draw this line in the figure above.*

Explanation. Slope of tangent line:

$$\left[\frac{dy}{dx}\right]_{(1,1)} = \frac{\pi \cdot (1) \cdot \sin(\pi(1) \cdot 1) - 1}{1 - \pi \cdot (1) \cdot \sin(\pi(1) \cdot (1))} = \frac{\pi \sin(\pi) - 1}{1 - \pi \sin(\pi)} = \frac{-1}{1} = -1$$

Equation of tangent line:

$$y - 1 = -1(x - 1) \implies y = -x + 2$$

Graph of tangent line: